Domain Adaptive and Interactive Differential Attention Network for Remote Sensing Image Change Detection

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***Abstract*— The objective of change detection (CD) is to identify the altered region between dual-temporal images. In pursuit of more precise change maps, numerous state-of-the-art (SOTA) methods design neural networks with robust discriminative capabilities. The convolutional neural network (CNN)-transformer model is specifically designed to integrate the strengths of the CNN and transformer, facilitating effective coupling of feature information. However, previous CNN-transformer studies have not effectively mitigated the interference of feature distribution differences as well as pseudovariations between two images due to cloud occlusion, imaging conditions, and other factors. In this article, we propose a domain adaptive and interactive differential attention network (DA-IDANet). This model incorporates domain adaptive constraints (DACs) to mitigate the interference of pseudovariations by mapping the two images to the same deep feature space for feature alignment. Furthermore, we designed the interactive differential attention module (IDAM), which effectively improves the feature representation and promotes the coupling of interactive differential discriminant information, thereby minimizing the impact of irrelevant information. Experiments on four datasets demonstrate the superior validity and robustness of our proposed model compared to other SOTA methods, as evident from both quantitative analysis and qualitative comparisons. The code will be available online (https://github.com/Jyl199904/DA-IDANet).**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), domain adaptation, interactive differential attention module (IDAM), remote sensing (RS), transformer.**

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I. Introduction

CHANGE detection (CD) seeks to recognize the alterations between the same geographical areas of dual-temporal images, ultimately generating a binary change map. In recent decades, remote sensing (RS) has robust capabilities of surface coverage and time-series observations, and combined with CD techniques has been successfully applied in various fields, encompassing urban expansion [1], deforestation [2], damage assessment [3], and forest cover mapping [4].

The approaches of CD can be broadly categorized into two main groups: traditional approaches [5] and deep learning approaches [6]. Traditional CD methods primarily utilize techniques, such as statistical analysis [7], signal processing [8], or machine learning [9]. When confronted with RS images exhibiting complex features or mixed pixels, traditional methods are limited by their modeling capabilities, making it difficult to efficiently identify changing regions [10]. Moreover, traditional methods typically depend on manual feature extraction or threshold setting, a time-consuming, and subjective process, especially when handling large-scale [11]. In recent years, it has become increasingly convenient to obtain high-resolution (HR) images. Nevertheless, there are rich texture information and complex spatial structures of ground objects in HR images, and traditional methods do not adequately capture complex and abstract features of images. To overcome the limitations of traditional methods, many researchers have turned to advanced techniques, such as deep learning, to address the CD task in an end-to-end pattern.

Deep learning models, with their capacity to capture and model complex nonlinear relationships, find extensive applications in the field of CD using RS images, showcasing notable effectiveness [12]. As the mainstream technology of deep learning, convolutional neural networks (CNNs) possess formidable feature extraction capabilities, enabling the automatic extraction of multilevel features containing rich spectral information and spatial features from RS images [13], [14], [15]. For instance, Wiratama et al. [16] devised a novel CNN-based model that amalgamates information from

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neighboring pixels in the image, facilitating the identification of altered regions through the interconnection of local features across various levels. Zhan et al. [17] crafted a weight-sharing Siamese deep CNN for pixelwise feature extraction and constructed feature vectors around altered pixel pairs, accentuating the regions of change. Chang et al. [18] proposed a multiscale joint information extraction method based on space and spectrum, which can accurately segment multitemporal image endmembers for CD. The existing deep learning methods often face challenges related to optimization problems with sparse constraints. Consequently, Feng et al. [19], [20] proposed a semisupervised CNN converted to reinforcement learning framework and a band-attention-based graph convolutional network for hyperspectral band selection, respectively. To fully utilize the complementary information between dual-temporal images and solve the problem of interscale variations, Du et al. [21] combined the discriminator network with CD network in an end-to-end manner. Huang et al. [22] designed a spatiotemporal augmentation and interlayer fusion network to improve the feature representation of the changing objects, thereby efficiently explore the information of the dual-temporal differences.

CNN models capture the spatial distribution and structure of ground objects in images through convolutional operations, which can progressively abstractly represent the information in the image layer by layer. However, the CNN model is constrained by a fixed receptive field and limited comprehension of deeper level information. This limitation can lead to an inadequate response to changes in ground objects at different scales and challenges in distinguishing certain complex changes in ground objects [23]. The attention mechanism can assist the model in concentrating on dynamic regions to augment feature representation [24]. With the incorporation of the attention mechanism, models can more effectively concentrate on areas with important information in the images [25]. It adaptively learns the importance of different parts of the original input images, thereby alleviating the constraint of fixed receptive fields inherent in CNN-only approaches. Hence, in recent years, many researchers have incorporated attention mechanisms into CD tasks to enhance accuracy. For instance, Liu et al. [24] devised feature exchange and channel attention modules to effectively model contextual information in dual-temporal images. Feng et al. [26] introduced a joint attention module by combining self-attention and cross-attention, which guides the global feature distribution on the input side and facilitates the coupling of information.

Recently, transformer [27], initially designed to address natural language processing tasks, has sparked a considerable sensation in the field of computer vision. In comparison to models solely based on CNN, transformer showcases robust capabilities in modeling long-range dependencies [28], [29], [30], which can capture global contextual information and process long-range associated features. This development has significantly propelled advancements in CD algorithms [31], [32], [33], [34]. For instance, Chen et al. [35] introduced a dual-temporal images transformer, employing an encoder to capture spatiotemporal context and refining original features through a decoder, which facilitates the efficient utilization of spatial contextual information and modeling, markedly enhancing the efficacy of CD. Zhang et al. [33] devised a U-shaped structure for the transformer framework, leveraging the Swin transformer as the basic unit. The model processes dual-temporal images through the encoder to extract multiscale features, concurrently restoring the details of change information through the decoder, which can obtain more accurate CD results.

In contrast to the straightforward CNN model that relies on convolutional layers to adeptly capture local features, the transformer model has the ability to concurrently attend to various segments of the input sequence through the multihead self-attention mechanism. This characteristic proves beneficial for global context modeling. Constructing a joint CNN-transformer model allows for a comprehensive amalgamation of the strengths of both architectures, effectively addressing multiscale changes in ground objects. This model excels in capturing local details while simultaneously considering global correlations [36], [37], [38]. Li et al. [39] introduced the ConvTrans block, which dynamically aggregates global features from the transformer module and local features from the CNN. This approach effectively enhances the robustness of CD at different scales, but it exhibits suboptimal performance when applied to extremely small altered areas. Chen et al. [40] presented a fully connected CNN-transformer, connecting UNet3+ and transformer in a Siamese structure. This model utilized a progressive attention module to deeply integrate features extracted by both the CNN and transformer, thereby addressing the issue of information loss between scales in CD. While the CNN-transformer model has demonstrated excellent performance in CD tasks, it still exhibits certain shortcomings. First, the increasing complexity of ground objects in bitemporal HR images, coupled with stylistic variations, such as illumination and cloud occlusion, may result in inconsistent feature representations for the same objects. Previous CNN-transformer models struggle to effectively mitigate the interference of the pseudovariations caused by differences in feature distributions. Furthermore, the existing methods are frequently difficult to sufficiently leverage the interactive differential discriminant information between input images, leading to ambiguous or inaccurate detection of ground objects and building boundaries.

To effectively address these challenges, we propose a domain adaptive and interactive differential attention network (DA-IDANet). Specifically, DA-IDANet first extracts multilevel features by a CNN backbone and employs domain adaptive constraints (DACs) mapping the input images to the same feature space for feature alignment. Then, a transformer module undertakes the encoding and decoding stages to aggregate context information. To fully leverage the interactive differential discriminant information between input images at different moments, an interactive differential attention module (IDAM) is proposed in DA-IDANet. The contributions of our work can be summarized as follows.

1) We have devised a hybrid CNN-transformer model that effectively integrates local–global features and spatial

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context information, thereby enhancing the accuracy of CD tasks.
2) We formulate DACs to mitigate the interferences of pseudovariations and feature distribution difference by mapping the two images to the same deep feature space for feature alignment.
3) The IDAM is designed to effectively integrate interactive differential discriminant information between the two images at different moments, thereby preserving crucial information of changing areas and maximizing the accuracy of boundary detection.

The remainder of this article is organized as follows. Section II provides a brief review of related work. The details of our proposed framework are elucidated in Section III. Section IV conducts a comprehensive experimental evaluation. The final section, Section V, summarizes this article.

II. Related Work

Recently, with the ongoing evolution of deep learning in CD, most mainstream algorithms are mainly based on feature learning. From the conventional CNN models extracting local features of images and performing fusion process to the subsequent introduction of attention mechanism to enhance feature representation. Then, transformer was proposed to capture global features and establish long-distance dependencies in feature representations. This section will delve into these aspects, considering the work from the perspectives of CNN models, attention mechanisms, and transformer.

A. CNN-Based Method

CNN-based models can leverage deeper neural network structures and incorporate information of different scales, which effectively capture the local spatial and texture details of RS images. Notably, the residual network [41] facilitates the direct transfer of information across layers through the introduction of residual connections. This mechanism increases network depth and effectively mitigates the vanishing gradient problem. Ding et al. [42] introduced a deep supervision module to enhance the capability of middle and deep layers in extracting more salient features. To better utilize the effective feature information in the deep network, Zhang et al. [43] proposed a deep supervised fusion network, employing a fully convolutional dual-stream architecture to extract highly representative deep features and then input into a deep differential discriminant network for CD. Li et al. [44] differentiated the changing areas of the target by leveraging deep contextual features. To mitigate the limited image rotation capability and exploit the unique correlation between image bands, Mei et al. [45] proposed cyclic polar coordinate convolutional layers to deal with rotational invariance for feature learning. In addition, Ma et al. [46] proposed a diverse band selection algorithm based on spectral correlation for effective image classification. Numerous experiments have substantiated the effectiveness of multiscale feature fusion [43], [47], [48]. This not only aids the model in adapting to changes at different scales but also enables the model to possess a larger receptive field. Consequently, the model that constructs multiscale feature fusion is better equipped to distinguish complex surface structures and changing areas in RS images. Lv et al. [49] integrated cross-layer blocks to fuse multiscale features and multilevel information based on UNet, thereby augmenting the model's ability to capture feature information. Peng et al. [50] introduced UNet++, which produced feature maps with higher spatial precision by incorporating global and fine-grained information. Shuai et al. [51] introduced a method utilizing graph convolutional operations to extract multiscale neighborhood features of nodes and then concatenated them through the attention mechanism.

It is noteworthy that CNN, being adept at capturing local details and small differences, plays a crucial role in extracting multiscale feature information essential for CD tasks. However, CNN lacks the ability to model long-range dependencies and contextual information interactions. Therefore, after the CNN extracts local features, the combination of the transformer architecture facilitates the establishment of long-distance spatial dependencies and contextual information interactions to improve the feature representation.

B. Attention-Based Method

In CD, the changing areas within input images are frequently concentrated in localized regions, while most areas remain static. Moreover, buildings or ground objects in the input images may be subject to various interferences, such as lighting and noise. Through the incorporation of the attention mechanism, the model can allocate more attention to areas that are particularly sensitive to changes, thereby reducing the processing of redundant information. This enhancement contributes to the overall robustness of the model. In recent years, the attention mechanism has been integrated into numerous studies of CD tasks. These mechanisms include channel attention [47], [52], spatial attention [25], [47], channel and spatial attention [43], [48], and so forth. To effectively improve the multiscale feature representation, Wang et al. [53] proposed the adaptive attention mechanism, effectively combines channel and spatial features to capture changes at various scales. Ding et al. [42] introduced cross-layer addition and skip connection modules guided by spatial attention to aggregate multilevel contextual information and direct the network's attention to changing areas. As the neural network's layers deepen, there is frequently an associated partial loss of information. Fang et al. [47] incorporated channel attention to enhance intermediate feature representation for modeling contextual information. Zhang et al. [43] employed the attention module to fuse deep features across diverse scales within the initial image and incorporate differential features to reconstruct the change map. To enhance the feature representation by exploiting the dependency between channel and spatial location, Liu et al. [54] introduced a paired attention module to strengthen the mutual influence between channels and spatial positions. Therefore, designing a suitable attention mechanism can improve the performance and efficiency of the model.

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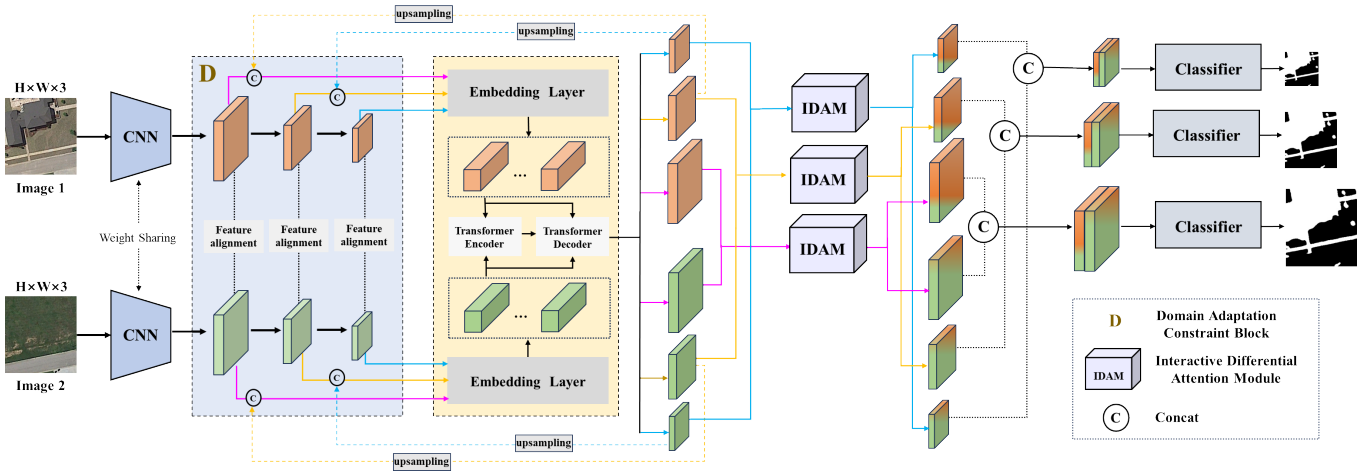

Fig. 1. Architecture of the proposed DA-IDANet. Bitemporal image features are extracted for the pretrained ResNet-18 and feature alignment is performed by domain adaptive constraints. Then, the contextual information is aggregated by the transformer encoder and decoder, and the interaction difference attention module is designed to supplement the interaction difference discriminative information between images. Finally, the final binary variation map is obtained by going through the classifier.

C. Transformer-Based Method

Transformer [27], introduced in 2017 as a novel architecture, has garnered attention for its ability to model remote dependencies, making it particularly impactful in processing sequence data. Due to its exceptional performance, researchers have extended the application of transformer to the realm of computer vision, yielding impressive results in various tasks, such as target detection [55], [56], superresolution [57], [58], semantic segmentation [47], [59], image classification [60], target tracking, video segmentation [55], and more. Recently, transformer has proven to have some growth potential in CD [33]. With transformer's advantage in long-range dependency modeling, which can help the model learn more discriminative global features, Chen et al. [35] proposed that the dual-temporal image transformer can effectively model context in the spatiotemporal domain and feedback the learned context information to the pixel space. Mei et al. [61] designed group-aware hierarchical transformer to concentrate the extracted global features for efficient image classification. Yan et al. [62] enhanced global view feature extraction using the Swin transformer architecture and combined multilevel visual features in a pyramid manner. To further enhance the global dependence of multiscale features and maximize the effective utilization of spatial information within refined features, Ke and Zhang [63] presented a hybrid multiscale transformer module, utilizing a detailed self-attention mechanism to model representation attention across mixed scales within each image. Simultaneously, Song et al. [64] introduced a multiscale Swin transformer combined with a deep supervision network, which incorporated deep supervision aggregation to enhance the distinguishability of multiscale features and fully utilized the refined multiscale spatial information. Models integrating CNN and transformer enable the harnessing of the strengths of both synthetically, fostering effective coupling of feature information. Feng et al. [65] introduced an intrascale cross-interaction and interscale feature fusion network to fully harness the potential of integrating CNN and transformer in a parallel manner. Liu et al. [66] introduced a CNN-transformer framework for multiscale upper and lower information aggregation, utilizing CNN to extract local features from images and decoding context information through the transformer multiscale aggregator, which can effectively identify the changing areas. Nevertheless, earlier CNN-transformer models lack the emphasis on interactive differential discriminant information between two images captured at distinct moments. Furthermore, they struggle to effectively alleviate the interference of pseudovariations and disparities in feature distribution resulting from factors, such as light irradiation, cloud occlusion, and other external influences. Therefore, we introduce the DACs and IDAM designed to align features extracted from dual-temporal images into a common feature space, fully harnessing the interactive differential discriminant information between them. Ultimately, it is possible to achieve effective mitigation of feature distribution differences and improve the accuracy of model detection boundaries.

III. Methodology

A. Architecture Overview

The proposed DA-IDANet is depicted in Fig. 1. Specifically, we input dual-temporal images into the pretrained ResNet-18 for feature extraction, and three different scales are selected to circumvent redundancy among information of features and diminish the computational load. Subsequently, the extracted features from the two images are aligned at each scale to the same deep feature space through DACs, thereby mitigating the interference of pseudovariations and feature distribution differences. Following this, the multiscale feature map undergoes the transformer encoder and decoder for the aggregation of context information. Then, the intermediate features are fed to the IDAM, utilizing the interactive differential discriminant information between input images to better focus on the changing areas and reduce the interference of irrelevant information. Finally, the feature maps

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are concatenated along the channel dimensions and passed through the classifier to obtain the final change binary map.

B. Multiscale Feature Extraction Network

DA-IDANet utilizes ResNet-18 [41] as the backbone network framework for extracting multiscale RS image features. The feature extraction module primarily comprises a 7×7 convolutional layer and four residual layers. A stride of 2 is employed with the 7×7 convolutional layer, resulting in a change of channels from 256 to 64. Simultaneously, it extracts the shallow features of the original images. Following this, the feature maps undergo a 3×3 max pooling layer, aiming to decrease the size of feature maps while preserving salient features. This strategy helps reduce the computational complexity and number of parameters in subsequent neural network layers, thereby enhancing computational efficiency.

In the backbone network, each residual block comprises two 3×3 convolutional layers, two batch normalization layers, and a rectified linear unit function. The convolutional layer is primarily employed to extract image features, and skip connections are introduced to directly add the input and convolutional results, ensuring consistency in the features. Both the second residual layer and the third residual layer set the stride to 2, which will cause the size of the feature map to be reduced by half. After the fourth residual block, the size of the final feature map is 1/8 of the input image size. This structure facilitates the gradual extraction of deep features from the image and diminishes the spatial size of the feature map for subsequent processing and analysis.

C. Domain Adaptation Constraint Block

The objective of domain adaptation is to apply knowledge learned from one or more domains to another domain by mapping it into a unified feature space [67]. Maximum mean discrepancy (MMD) is a common domain adaptation technique, which fosters feature alignment between different domains by measuring the distribution distance of two random variables between the source and target domains. Constructing the MMD loss can maximize the similarity of feature representations in the feature space, thereby reducing the distribution difference between the features of the source and target domains. While the two images originate from a unified sensor, the influence of external factors, such as light irradiation, cloud cover, and imaging conditions, can lead to variations in spectral curves and differences in feature distribution within the same category of features. By introducing DACs, images from two different moments are aligned to the same deep feature space, thereby reducing the interference caused by feature distribution differences as well as pseudovariations. The MMD is defined as

$$\mathrm{MMD}\big[F, p, q\big] := \sup\big(E_p\big[f(x)\big] - E_q\big[f(y)\big]\big) \tag{1}$$

where $E_P(\cdot)$ is expressed as expected value, sup is expressed as supremum, and $f(\cdot)$ is expressed as a mapping function. The function domain in MMD was defined as any vector f within the unit sphere of the regenerated Hilbert space (i.e., $\|f\|_{\mathcal{H}} \leq 1$), which means that the norm of the base vector f in the regenerated Hilbert space should be less than or equal to 1. The function $\phi(\cdot)$ corresponding to the kernel function maps the random variable x to infinite dimensions ($\phi(\cdot)$), and $f(x)$ represented the dot product between the base vector f and $\phi(x)$ in the regenerated Hilbert space.

Initially, expand the expectation and subsequently express the function $f(x)$ in the Hilbert space as the inner product of the mapping function $\phi(\cdot)$ and the basis vector f. This means that mapping the vector $f(x)$ to the Hilbert space through the function $\phi(\cdot)$, followed by performing an inner product with the base vector f within a unit ball in the space. This process accomplishes the transformation to higher dimensions. Consequently, it can be inferred as follows:

$$\begin{aligned} E_p\big[f(x)\big] &= \int_{\mathcal{X}} p(dx) f(x) \\ &= \int_{\mathcal{X}} p(dx) \langle \phi(x), f \rangle \\ &= \left\langle \int_{\mathcal{X}} p(dx) \phi(x),\, f \right\rangle. \end{aligned} \tag{2}$$

Combined with the basic definition, the following expression can be obtained:

$$\begin{aligned} \mathrm{MMD}[p, q, H] &:= \sup_{f \in H, \|f\|_H \leq 1} \left(E_p[f(x)] - E_q[f(y)]\right) \\ &\leq \sup_{f \in H, \|f\|_H \leq 1} \left\|\mu_p - \mu_q\right\|_{\mathcal{H}} \|f\|_{\mathcal{H}} \\ &\leq \left\|\mu_p - \mu_q\right\|_{\mathcal{H}} \end{aligned} \tag{3}$$

where $\mu_p = \int_{\mathcal{X}} p(dx)\phi(x)$ represents the kernel mean embedding. Assume that the source-domain data and target-domain data amounts are m and n, respectively, the final formula can be obtained as follows:

$$\begin{aligned} \mathrm{MMD}[p, q, H] &= \left\|\mu_p - \mu_q\right\|_{\mathcal{H}} \\ &= \left\| \frac{1}{n} \sum_{i=1}^{n} \phi(x_i) - \frac{1}{m} \sum_{j=1}^{n} \phi\big(y_j\big) \right\|_{\mathcal{H}}. \end{aligned} \tag{4}$$

D. Interactive Differential Attention Module

The introduction of the attention mechanism aids the model in more effectively pinpointing potential change areas and enhances the model's perceptual capabilities, ultimately improving the accuracy of CD. Usually, with the increasing number of layers in the neural network, the model tends to lose certain details from the original images. Additionally, the changing areas might constitute only a small portion of the entire image and exhibit sparse distribution, which may result in blurred or inaccurate boundary detection. Therefore, we propose the IDAM to effectively supplement the learning of interactive differential discriminant information between the input images. With the IDAM, the regions associated with change can be dynamically focused on, thus reducing the interference of irrelevant information and maximizing the accuracy of boundary detection.

Built upon the concept of the traditional self-attention mechanism, the IDAM attains global context awareness by modeling dependencies between various positions in the sequence. The feature map is represented as a 3-D tensor

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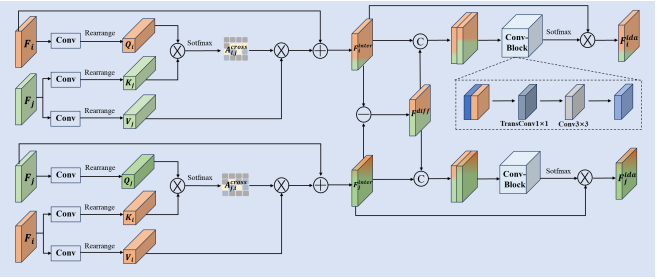

Fig. 2. Architecture of the proposed interactive differential attention module.

and then reshaped into a 2-D matrix. The self-attention mechanism incorporates three sets of weight matrices, which typically match the dimensions of the input features and the expected output dimensions. The attention score is computed by taking the dot product between Q and K, followed by dividing the result by a scaling factor to prevent gradient explosion. Throughout this process, an attention score matrix is generated for different positions, indicating the degree of correlation between each position and other positions. Ultimately, the attention score matrix is normalized into a probability distribution through the softmax operation to acquire the attention weight matrix. Subsequently, the V matrix is weighted and operated by the attention weight matrix to yield the output of the self-attention mechanism, as shown in formula (5). This output encapsulates information about the importance of each position in the overall sequence, facilitating global context awareness. This mechanism proves valuable in capturing long-distance dependencies and strengthening the model's representation capability

$$\text{Self_Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d}}\right) V_i \tag{5}$$

where the vector sequence $Q_i \in R^{N\times C}$, $K_i \in R^{N\times C}$, and $V_i \in R^{N\times C}$, where $N = H \times W$. While the self-attention mechanism effectively captures global context information by modeling dependence relationships at different locations, it exhibits certain limitations in capturing interactive differential discriminant information between features extracted from dual-temporal images. Therefore, we designed the IDAM to address the above limitations.

As depicted in Fig. 2, the features F_i and F_j of the two images extracted in the previous stage are, respectively, input into a 3×3 convolutional layer to obtain Q_i, K_j, and V_j, proceeding to execute the corresponding cross-attention mechanism operation

$$\begin{aligned} F_i'(Q_i, K_j, V_j) &= \text{Softmax}\left(\frac{Q_i K_j^T}{\sqrt{d_i}}\right) \cdot V_j \qquad (6)\\ &= A_{i,j}^{\text{cross}} \cdot V_j \\ F_i^{\text{inter}} &= F_i + F_i'(Q_i, K_j, V_j) \qquad (7) \end{aligned}$$

where i and j, respectively, represent the sequence numbers of the feature maps at different moments, and $A_{i,j}^{\text{cross}}$ represents the cross-attention probability distribution. We acquire the global attention distribution by normalizing Q and K of feature maps at different moments. Similarly, we also calculate $F_j'(Q_i, K_j, V_j)$ and F_j^{inter}. The interactive discriminative information between images enhances the sensitivity to surface change information, which further aids the model in effectively distinguishing the contextual information of the changed area and identifying the location of the change

$$F^{\text{diff}} = F_i^{\text{inter}} - F_j^{\text{inter}} \tag{8}$$

$$\begin{aligned} F_i'' &= \text{Conv}\left(\text{Concat}\left(F^{\text{diff}}, F_i^{\text{inter}}\right)\right) \\ F_j'' &= \text{Conv}\left(\text{Concat}\left(F^{\text{diff}}, F_j^{\text{inter}}\right)\right) \qquad (9)\\ F_i^{\text{ida}} &= F_i^{\text{inter}} \cdot \text{Softmax}\left(F_i''\right) \\ F_j^{\text{ida}} &= F_j^{\text{inter}} \cdot \text{Softmax}\left(F_j''\right). \qquad (10) \end{aligned}$$

As depicted in formula (8), we execute a differential operation between F_i^{inter} and F_j^{inter} to obtain the differential features F^{diff}. Subsequently, the differential features F^{diff} and the interactive features F_i^{inter} are concatenated on the channel dimension. Following a sequence of convolutional operations, the softmax and normalization functions are applied to obtain the probability distribution. The IDAM efficiently mitigates the issue of partial information loss caused by an increasing number of neural network layers during the training. It maximizes the utilization of interactive differential discriminative feature information between dual-temporal images, ultimately leading to enhancements in both the overall accuracy and robustness of the model.

E. Prediction Head and Loss Function

In the concluding phase of generating the binary changed map, a universal prediction head is employed. Initially, three decoded aggregated maps at different scales are upsampled to the same size as the input images with bilinear interpolation. Subsequently, these maps are converted into a differential prediction through two 1×1 convolutional layers and batch normalization. Following multiscale differential acquisition, aggregation, and reorganization, we generated three predictions (P_{cat}', P_{cat}'', and P_{cat}'''). During the training phase, we optimize the model parameters by minimizing the domain adaptation constraint loss L_{mmd} and cross-entropy loss L_{ce} between the three predictions and the ground truth one by one

$$P_{\text{cat}} = \text{Conv}(\text{conv}(\text{concat}(\text{up}(F^i da_j), (\text{up}(F^i da_j)))) \tag{11}$$

$$L_{\text{ce}}(Y, P) = -\frac{1}{N}\sum_{i=1}^{N}[Y_i \log_2 P_i + (1 - Y_i)\log_2(1 - P_i)] \tag{12}$$

$$L_{\text{mmd}}\left(F_i, F_j, H\right) = \left\| \frac{1}{n}\sum_{i=1}^{n}\phi(x_i) - \frac{1}{n}\sum_{j=1}^{n}\phi\left(y_j\right) \right\|_{\mathcal{H}} \tag{13}$$

where N represents the total number of pixels in the change map, n represents the total number of samples, Y_i is the label of sample i, P_i is the probability of sample i to be predicted as a positive class, σ represents the sigmoid function, GT is the ground truth, and λ is the weighting factors of DAC loss.

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The total training loss is given by the following formula:

$$L_{\text{total}} = \sum_{i=1}^{3} L_{\text{ce}}(\text{GT}, \sigma(P_i)) + \lambda L_{\text{mmd}} \quad (F_i, F_j, H). \tag{14}$$

In the ultimate inference stage, prediction P_{cat}''' was selected as the output result map and converted into a single-channel map through an argmax operation.

IV. Experiments

A. Data Description

To assess the practical performance of the proposed DA-IDANet, we conducted extensive experiments on four representative public RS CD datasets.

1) *CDD [68]:* The CDD dataset, sourced from Google Earth, is a public dataset focused on detecting seasonal changes in the same region. It comprises multispectral and HR images, with resolutions ranging from 0.03 to 1 m, typically obtained from satellites or aerial photography to cover a diverse geographical area and changes at different times. After cropping and rotating operations, 16 000 image patches of the size 256 × 256 pixels are derived from the original images, which were randomly split into 10 000/3000/3000 for training, validation, and testing.
2) *LEVIR-CD:* The learning, vision (LEVIR-CD) [1] is an extensive very HR (VHR, 0.5 m/pixel) CD dataset. It encompasses 637 pairs of image patches with a size of 1024 pixels, capturing significant changes over a span of 5–14 years. The dual-temporal images of LEVIR-CD are sourced from 20 distinct regions across various cities in TX, USA. For experimental purpose, we cropped them into smaller image pairs with a size of 256 pixels, and they were randomly divided into 7120/1024/2048 for training, validation, and testing.
3) *CLCD-CD [69]:* The Change in Land Cover Dataset (CLCD) comprises 600 pairs of farmland change images. Among these, they were randomly divided into 360/120/120 for training, validation, and testing. The spatial resolution ranges from 0.5 to 2 m. Each set of samples includes two 512 × 512 images and is accompanied by a binary tag indicating the change in farmland.
4) *WHU-CD [70]:* The Wuhan University dataset encompasses the region affected by the earthquake in February 2011, along with the area that underwent reconstruction several years after the earthquake. The dataset spans approximately 20 km^2 and includes 12 796 buildings. It comprises a pair of aerial images with a spatial size of 32 507 × 15 354 pixels with a resolution of 0.2 m/pixel. The original images were cropped into smaller image pairs with the size of 256 pixels and randomly divided into 6096/762/762 for training, validation, and testing.

B. Experimental Setup

1) Comparison With State of the Arts (SOTAs): To assess the effectiveness and accuracy of our DA-IDANet in CD, we compared with several SOTA approaches, which including FC-EF [14], FC-Siam-Di [14], FC-Siam-Conc [14], IFNet [43], SNUNet [47], BiT [35], DASNet [70], STANet [1], FTN [62], DMINet [26], and MSCANet [66].

1) FC-EF [14] is a CNN-based CD network built upon UNet, which takes dual-temporal RS images as the input layer of the network and fuses at the image level.
2) FC-Siam-Di [14] builds upon the improvement of FC-EF, extracting multilevel features from dual-temporal RS images and fusing them using the shared-weight characteristics of the Siamese network.
3) FC-Siam-Conc [14] further refines FC-EF by incorporating feature-level fusion through a Siamese network.
4) IFNet [43] leverages multiscale features from dual-temporal images and integrates the differential features obtained through the attention module to produce the ultimate binary change map.
5) SNUNet [47] is a densely connected Siamese network that enhances the feature representation in the middle layer and models context using the channel attention module.
6) BiT [35] models the spatiotemporal context of dual-temporal images using the transformer encoder, feeds back the learned context information, and refines the initial features through the transformer decoder.
7) DASNet [70] captures extended dependencies using a dual attention mechanism to acquire a more discriminative feature representation.
8) STANet [1] computes attention weights between pixels at diverse times and locations using a self-attention module, leveraging this information to generate a more discriminative feature representation.
9) FTN [62] proposes a progressive attention module and leverages Swin transformer to learn more discriminative global features by modeling long-range dependencies.
10) DMINet [26] utilizes the intertemporal joint attention to steer the global feature distribution while eliciting representational information coupling.
11) MSCANet [66] adopts the CNN-transformer structure, extracts multiscale features through the CNN backbone network, and then aggregates contextual information based on the transformer structure.

2) Experimental Details: The proposed DA-IDANet and all compared methods were experimented on an NVIDIA GeForce RTX 3080Ti workstation, using the PyTorch framework as the model development environment. Throughout the model training phase, we applied data augmentation techniques to the dataset, encompassing vertical and horizontal flipping, scaling, and cropping to mitigate overfitting during training. The model underwent training utilizing the adaptive moment estimation, employing a batch size of 8, λ was set as 0.4. The training spanned 100 epochs, and the initial learning rate was 1e-4.

3) Evaluation Metrics: To evaluate the performance and accuracy of our proposed model compared with other SOTA methods, we use five evaluation metrics to measure the similarity between the predicted change probability map and the ground truth. These metrics include precision (Pre), recall (Rec), F1 score (F1), intersection over union (IOU), and kappa

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TABLE I
COMPARISON RESULTS ON THE LEVIR-CD AND CDD DATASET. THE BEST RESULTS ARE HIGHLIGHTER IN BOLD. ALL RESULTS ARE EXPRESSED AS PERCENTAGE (%)

Methods	LEVIR-CD						CDD					
	Precision	Recall	F1	IOU	Kappa	OA	Precision	Recall	F1	IOU	Kappa	OA
FC_EF	87.90	62.93	73.35	57.91	71.85	97.08	81.73	53.36	64.57	47.67	60.22	92.90
FC_Siam_conc	90.78	64.83	75.64	60.82	74.27	97.22	91.32	53.15	67.19	50.59	63.47	93.50
FC_Siam_diff	90.58	62.20	73.50	58.42	72.32	97.17	90.75	52.19	66.27	49.56	62.48	94.56
IFNet	**93.64**	66.15	77.53	63.30	76.28	97.55	92.84	83.41	87.87	78.37	86.05	97.90
SNUNet	90.30	71.23	79.64	66.17	78.42	97.68	91.21	85.51	88.27	79.00	86.14	97.96
BiT	90.52	71.75	80.05	66.74	78.86	97.72	92.63	83.91	88.06	78.66	78.86	97.84
DASNet	91.18	68.17	78.01	63.95	76.74	97.55	91.76	85.34	88.43	79.23	85.66	98.15
STANet	91.35	65.69	76.42	62.12	75.32	97.44	92.26	84.49	88.20	78.90	86.21	97.84
FTN	91.05	72.67	80.82	68.18	80.09	97.75	**95.45**	77.18	85.35	74.45	83.08	98.04
DMINet	91.92	71.11	80.19	68.37	78.61	97.47	91.66	85.53	88.49	79.36	86.72	96.92
MSCANet	90.11	73.43	80.92	67.95	79.76	97.79	92.11	84.43	88.10	78.74	86.29	97.85
Ours	91.60	**73.73**	**81.70**	**68.82**	**80.52**	**98.12**	92.88	**87.05**	**89.87**	**80.56**	**87.19**	**98.23**

(Kap). The definition of each indicator is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{16}$$

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{18}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{19}$$

$$\text{Kappa} = \frac{\text{OA}-\text{P}}{1-\text{P}} \tag{20}$$

$$P = \frac{(\text{TP}+\text{FP})(\text{TP}+\text{FN})+(\text{FN}+\text{TN})(\text{TP}+\text{TN})}{(\text{TP}+\text{FP}+\text{TN}+\text{FN})^2} \tag{21}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. In kappa, P represents the hypothetical probability that the reference is consistent with the prediction.

C. Performance Comparison and Result Analysis

The quantitative comparison of CD indicators is presented in Tables I and II. The DA-IDANet demonstrates significant advantages over other SOTA models through five indicators. For instance, on the CDD dataset, compared to MSCANet, DA-IDANet exhibited improvements of 2.62% in Recall, 1.77% in F1 score, and 1.82% in IOU. Compared to FTN, which boasts superior precision owing to its progressive attention mechanism designed based on Swin transformer, DA-IDANet exhibits a 2.57% decrease in precision indicators. Nevertheless, DA-IDANet excels over FTN in various other performance metrics. On the CLCD-CD, DA-IDANet outperformed MSCANet by 3.48% in recall, 2.22% in F1 score, and 2.45% in IOU. However, compared to FTN, DA-IDANet dropped by 1.51% in precision. On the LEVIR-CD dataset, we improved the F1 score by 0.78% and the IOU by 0.87% compared to MSCANet. In comparison with IFNet, DA-IDANet experienced a 2.04% decrease in precision, suggesting that the deeply supervised image fusion design of IFNet yields superior precision benefits. Among other indicators, DA-IDANet maintains certain advantages. Similarly, on the WHU-CD dataset, we achieved improvements of 1.01% in recall, 0.85% in F1 score, and 1.28% in IOU compared to MSCANet. These numerical results provide clear evidence that the DAC and IDAM contribute meaningfully to CD tasks. Visualization results for each model on different datasets are presented in Figs. 3–6. These figures showcase the accuracy of different models in detecting changing areas using different colors to represent true positives (TP, white), true negatives (TN, black), false positives (FP, red), and false negatives (FN, green).

1) Visualization on LEVIR-CD: Fig. 3(a) and (d) depicts the relatively scattered small buildings, while Fig. 3(b) and (c) showcases the areas possibly affected by trees, with blocked large and small buildings. Fig. 3(e) and (f) presents the examples of house building renovations. In the detection of large buildings, the DA-IDANet consistently exhibits more accurate predictions of complete change maps. This demonstrates the effectiveness of the IDAM in addressing problems, such as boundary blurring, missed detections, and false detections that may occur during the detection processing. Importantly, our proposed model demonstrates robust performance in detecting buildings of various sizes, even in the presence of tree occlusion or light interference. The

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TABLE II
COMPARISON RESULTS ON THE CLCD-CD AND WHU-CD DATASET

Methods	CLCD-CD						WHU-CD					
	Precision	Recall	F1	IOU	Kappa	OA	Precision	Recall	F1	IOU	Kappa	OA
FC_EF	70.24	47.70	56.82	39.68	54.05	94.60	81.12	68.12	74.05	58.80	73.08	98.10
FC_Siam_conc	65.62	54.27	59.41	42.25	56.47	94.48	54.40	82.88	65.69	48.91	63.96	96.56
FC_Siam_diff	67.88	51.18	58.36	41.20	55.51	94.57	62.68	80.25	70.39	54.31	69.00	97.32
IFNet	75.02	63.49	68.77	51.92	66.01	95.62	90.58	86.99	88.75	79.77	88.29	99.12
SNUNet	66.67	65.09	65.87	49.11	63.16	94.98	76.25	85.09	80.43	67.26	79.57	98.35
BiT	72.85	71.59	72.21	54.51	68.01	95.90	86.69	89.39	88.02	78.60	87.52	99.03
DASNet	67.86	71.36	69.57	53.33	67.05	93.35	89.59	88.15	88.86	79.96	88.41	99.12
STANet	75.10	63.16	68.61	52.74	66.82	95.79	91.80	88.41	90.07	81.67	89.37	99.15
FTN	**76.73**	63.03	69.21	54.03	68.01	95.24	**92.54**	86.13	89.22	81.83	89.40	99.20
DMINet	74.65	72.40	73.51	56.12	69.42	96.12	88.33	89.59	88.96	79.23	87.92	99.06
MSCANet	74.36	70.66	72.46	54.77	68.43	95.66	90.20	89.11	89.65	81.25	89.23	99.18
Ours	75.22	**74.14**	**74.68**	**57.22**	**70.11**	**96.35**	90.88	**90.12**	**90.50**	**82.53**	**89.95**	**99.25**

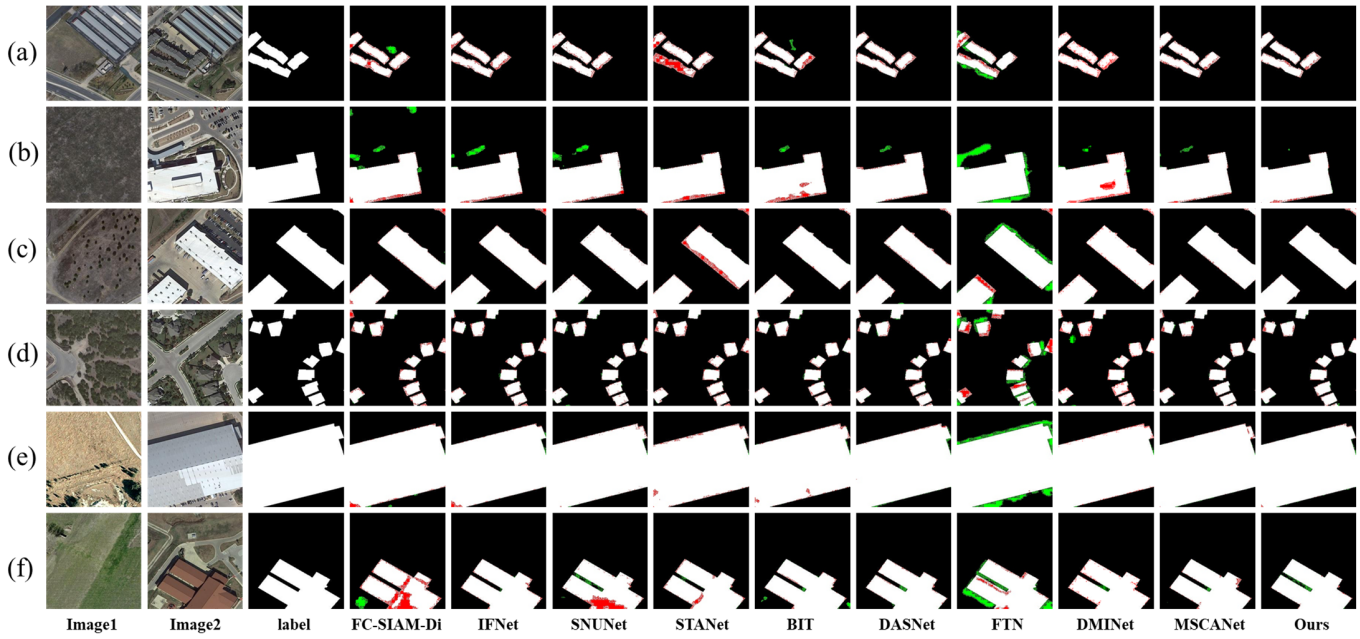

Fig. 3. Visual results of different models on test samples of LEVIR-CD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

visual comparison in Fig. 3, combined with the quantitative analysis of the evaluation metrics in Table I, underscores the leading performance of our proposed DA-IDANet on the LEVIR-CD dataset.

2) Visualization on CDD: These selected images depict changes in ground objects or buildings corresponding to seasonal variations. Fig. 4(a), (b), (d), and (f) illustrates the changes in buildings or roads occurring with the shifting seasons. Fig. 4(c) and (e) showcases different types of ground objects and the presence of small vehicles captured after seasonal changes. It is noteworthy that the model may be influenced by seasonal changes or severe weather conditions during CD tasks, such as occlusion by lawns or snow cover. Other methods might miss or incorrectly identify roads or buildings, and edge detection for medium and large buildings may lack clarity. However, with the incorporation of DACs, the features extracted from dual-temporal images can be aligned to a common feature space. This feature alignment effectively mitigates the interference of pseudovariations arising from differences in feature distributions. Consequently, the DA-IDANet demonstrates a more accurate detection of changing areas.

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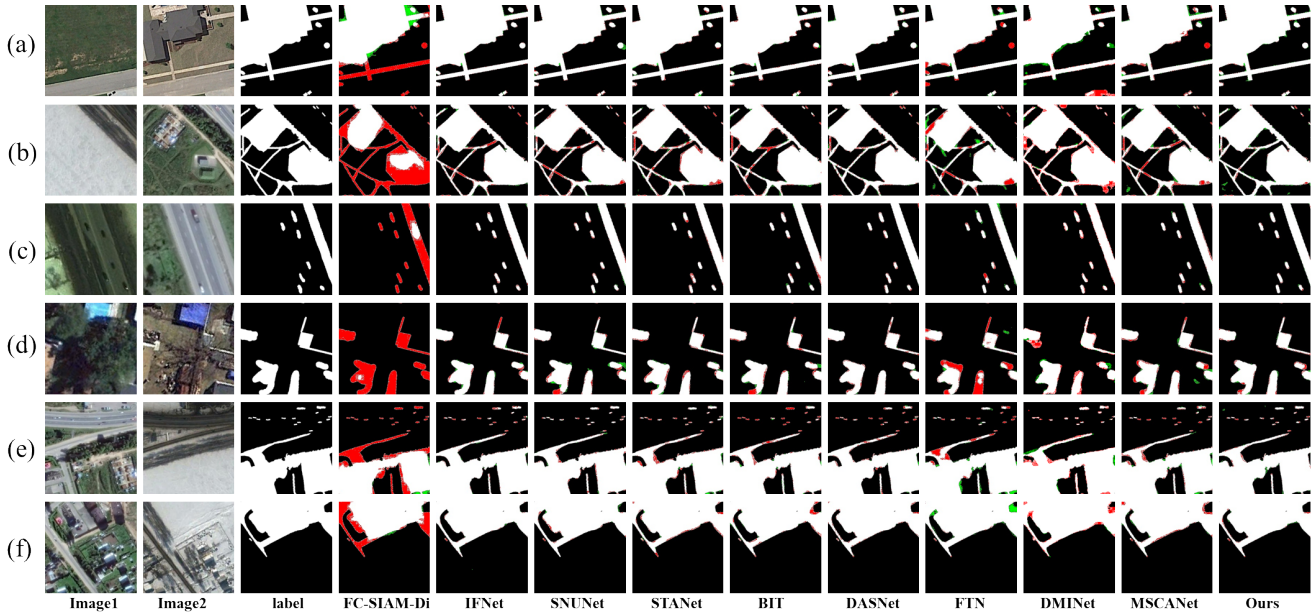

Fig. 4. Visual results of different models on test samples of CDD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

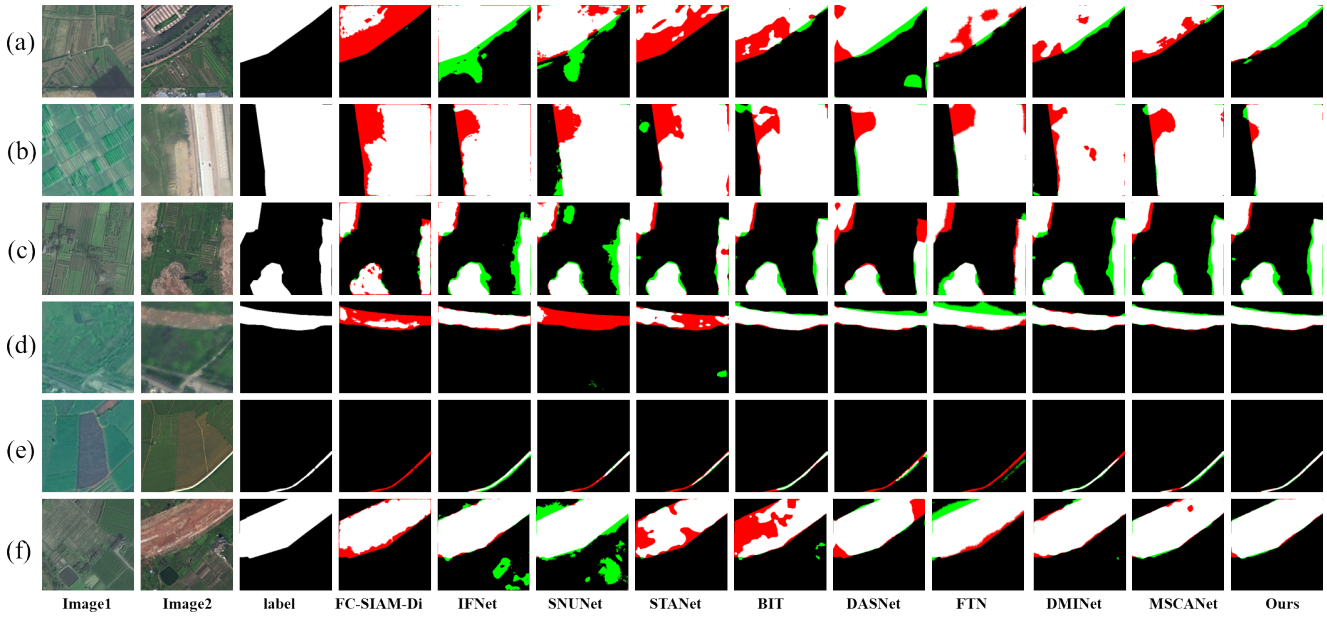

Fig. 5. Visual results of different models on test samples of CLCD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

3) Visualization on CLCD: The CLCD dataset encompasses a variety of changes in farmland and other land features. Fig. 5(a)–(f) illustrates the changes in buildings on farmland, modifications to roads, changes in farmland, and alterations in rivers, emphasizing the model's crucial accuracy in detecting these changes. Accurately identifying changing areas in complex farmland and other land features poses a significant challenge. Based on both visual renderings and the insights from Table II, our proposed model demonstrates superior accuracy in detecting changes in such intricate environments compared to other models, including the ability to discern more complete boundaries.

4) Visualization on WHU-CD: Fig. 6(a), (e), and (f) depicts the areas with relatively dense or small housing buildings, while Fig. 6(b)–(d) showcases the instances of greenhouse construction and asphalt-repainted roads. Examining several pairs of representative cases visually, where architectural change areas are small or dense, poses a challenge for CD tasks. Our DA-IDANet demonstrates a lower number of FPs and FNs compared to other models, achieving more accurate and efficient change maps.

D. Ablation Analysis

Ablation experiments were conducted to assess the effectiveness and rationality of each proposed module by

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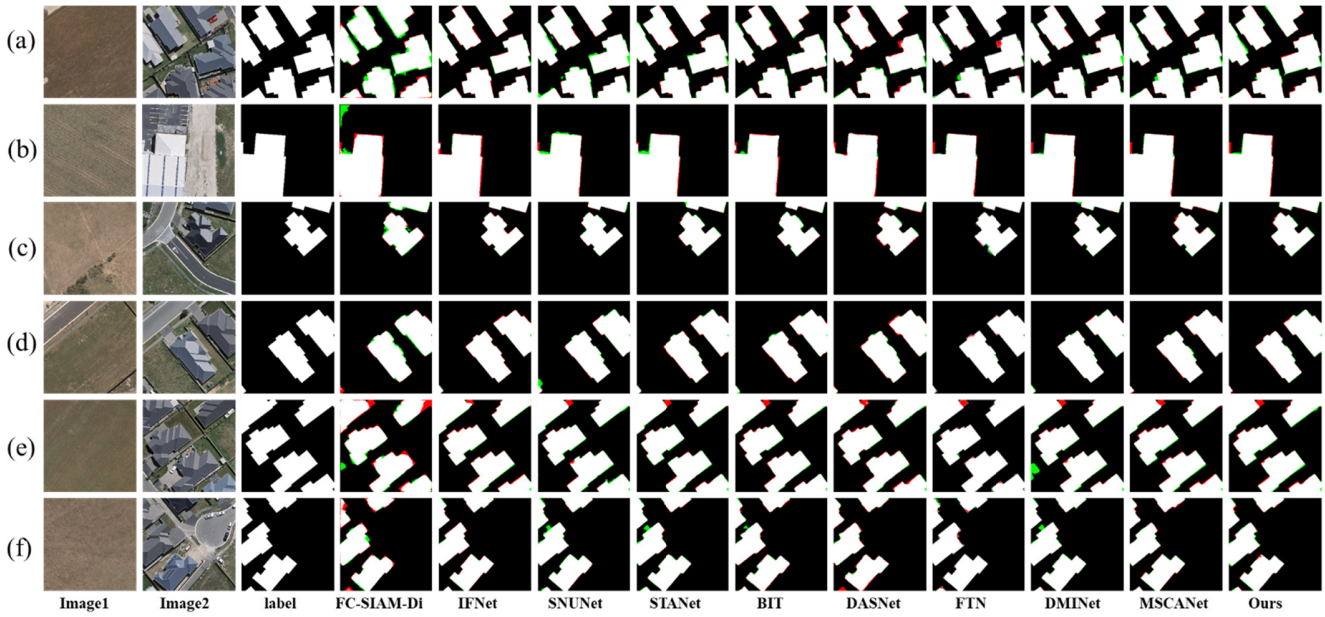

Fig. 6. Visual results of different models on the test samples of WHU-CD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

TABLE III

ABLATION RESULTS (%) ON CD DATASET. THE SYMBOL "X" INDICATES THAT THE CORRESPONDING MODULE WAS REMOVED OR OPERATION WAS NOT PERFORMED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Change IDAM Position	Change DAC Position	DAC	IDAM	Multiscale	LEVIR-CD		CDD		WHU-CD		CLCD	
						F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)
DA-IDANet	×	×	×	×	√	80.92	67.95	88.10	78.74	89.65	81.25	72.46	54.77
	√	×	√	√	√	81.17	68.23	88.85	79.94	89.75	81.41	72.76	55.96
	×	√	√	√	√	80.80	67.82	88.33	79.10	89.34	81.12	72.51	55.36
	×	×	√	√	×	79.03	65.33	87.68	78.06	89.63	81.22	67.38	50.81
	×	×	√	×	√	80.97	68.11	88.27	79.00	89.84	81.52	72.79	56.22
	×	×	×	√	√	81.29	68.46	88.91	80.03	89.92	81.84	73.52	56.37
	×	×	√	√	√	**81.70**	**68.82**	**89.87**	**80.56**	**90.50**	**82.53**	**74.68**	**57.22**

swapping their positions or removing them from the CD network. Table III provides the experimental results on four datasets.

1) DACs: To assess the impact of DACs on model performance, we conducted experiments by removing the DACs. When DACs are removed, the F1 scores of the model dropped significantly, with 81.29% on LEVIR-CD, 88.91% on CDD, 89.92% on WHU-CD, and 73.52% on CLCD, respectively. It can fully illustrate that DACs play a certain role in the feature alignment of the model.

In addition, to evaluate the reasonableness of the location of the domain adaptation constraints, we placed the constraint after the transformer aggregates the context information rather than after the feature extraction network. According to the result in Table III, the F1 scores of the modified model on LEVIR-CD, CDD, WHU-CD, and CLCD are 80.80%, 88.33%, 89.34%, and 72.51%, respectively. It is evident that changing the position of the DACs, compared with the original solution, caused a drop in accuracy. This indicates that higher benefits can be obtained by conducting feature alignment after the CNN extracts multiscale features.

2) IDAM: To precisely assess the overall impact of the IDAM on model performance, we removed the IDAM. The result in Table III reveals that the F1 scores of the model on the LEVIR-CD, CDD, WHU-CD, and CLCD are 80.97%, 88.27%, 89.84%, and 72.79%, respectively. After removing the

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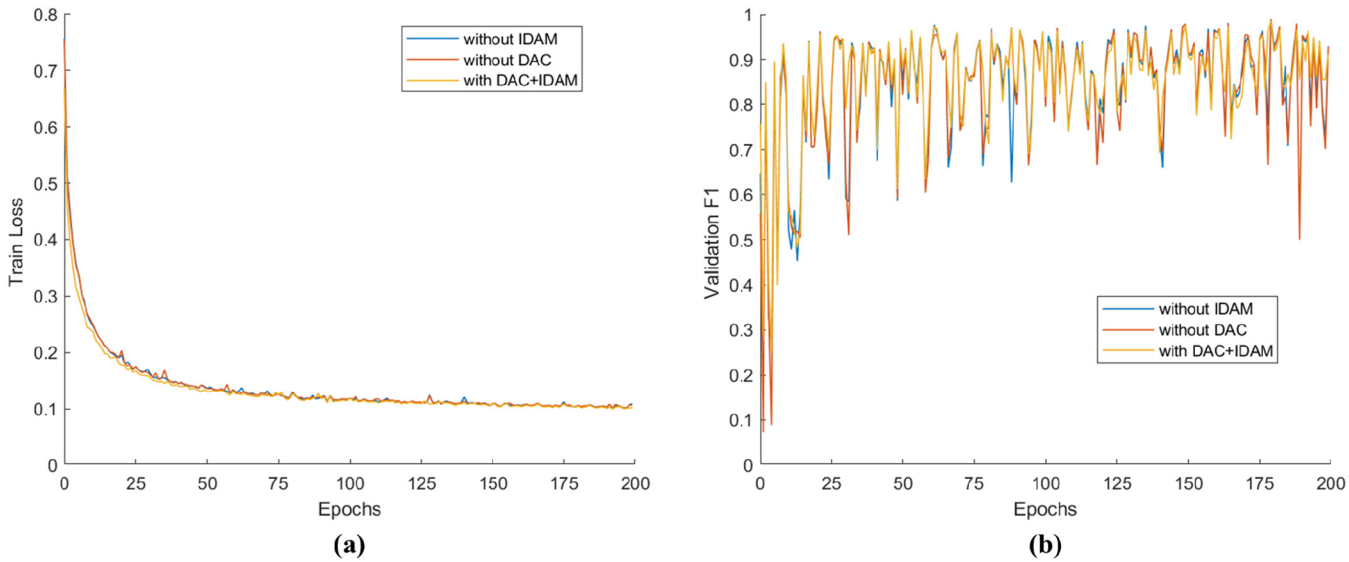

Fig. 7. (a) Training loss changes with increasing training epoch. (b) Validation F1 changes with increasing training epoch.

IDAM, the attention to the interactive differential discriminant information between the two images would be significantly reduced, thereby affecting the performance of the model.

Additionally, we compared the difference in performance of the IDAM at different locations. We attempted to position the IDAM before the transformer aggregates contextual information, but the effectiveness was not as significant as the original design solution. It can be observed that the F1 scores of the model on the LEVIR-CD, CDD, WHU-CD, and CLCD are 81.17%, 88.85%, 89.75%, and 72.76%, respectively. The rate is lower than the original scheme by 0.52%, 1.02%, 0.75%, and 2.22%. From this, it can be concluded that utilizing the IDAM after aggregating contextual information through transformer helps the model better identify the changed areas.

3) Multiscale features: To verify the influence of the introduced multiscale features, we removed the features of two scale. The result in Table III indicates that removing the features of two scale caused a decrease in F1 scores on the LEVIR-CD, CDD, WHU-CD, and CLCD by 2.66%, 2.19%, 0.83%, and 7.3%, respectively. Additionally, compared to the original solution, the IOU fell by 3.49%, 2.5%, 1.31%, and 6.41%, respectively. Smaller scale features aid in detecting small changes in ground features, while larger scales are effective in capturing changes in the overall ground feature structure. Combining multiscale information allows for a more comprehensive and accurate representation of surface changes. Simultaneously, we conducted a comparison of the variation in loss during the training process and F1 scores during the validation process, as illustrated in Fig. 7. The loss of complete model can converge faster and which has more stable F1 scores. This suggests that DA-IDANet exhibits a better performance of generalization.

E. Model Efficiency Assessment

To compare the operational efficiency of various models, we evaluate the computational complexity and space complexity using FLOPs and Params for input image sizes of 256 × 256 pixels and 512 × 512 pixels. As shown in Table IV, FC-EF, FC-Siam-Conc, and FC-Siam-Di have lower FLOPs and Params compared to other models at different input image sizes. This is attributed to their basis on traditional CNN structures without the incorporation of complex network architectures, resulting in lower computational and space complexity. IFNet and DASNet exhibit higher FLOPs and Params compared to other models. This is attributed to their relatively complex attention calculation mechanism, leading to a larger number of network parameters. The proposed DA-IDANet has 15.45G FLOPs and 16.45M Params when the input image size is 256 pixels. This indicates that our proposed model can maintain excellent CD performance while also

TABLE IV
EFFICIENCY COMPARISON ANALYSIS WITH DIFFERENT INPUT SIZES

Methods	Image size:256			Image size:512	
	Average F1(%)	Flops(G)	Params (M)	Flops (G)	Params (M)
FC_EF	67.20	3.58	1.35	14.30	1.35
FC_Siam_conc	66.98	5.33	1.55	21.32	1.55
FC_Siam_diff	67.13	4.73	1.35	18.91	1.35
IFNet	80.73	82.26	35.73	329.03	35.73
SNUNet	78.55	54.83	12.04	219.34	12.04
BiT	82.09	8.75	3.49	42.37	3.49
DASNet	81.22	107.98	55.31	432.00	55.31
STANet	80.83	18.07	16.93	72.30	16.93
FTN	81.15	44.42	164.45	178.57	164.45
DMINet	82.78	14.55	6.24	59.49	6.24
MSCANet	82.79	14.80	16.42	59.08	16.42
Ours	84.19	15.45	16.45	61.65	16.45

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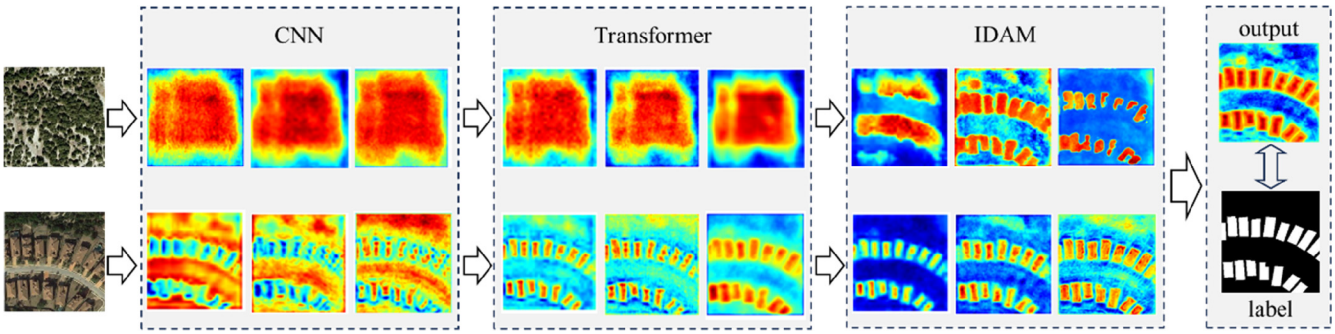

Fig. 8. Network visualization taking images from LEVIR-CD as examples.

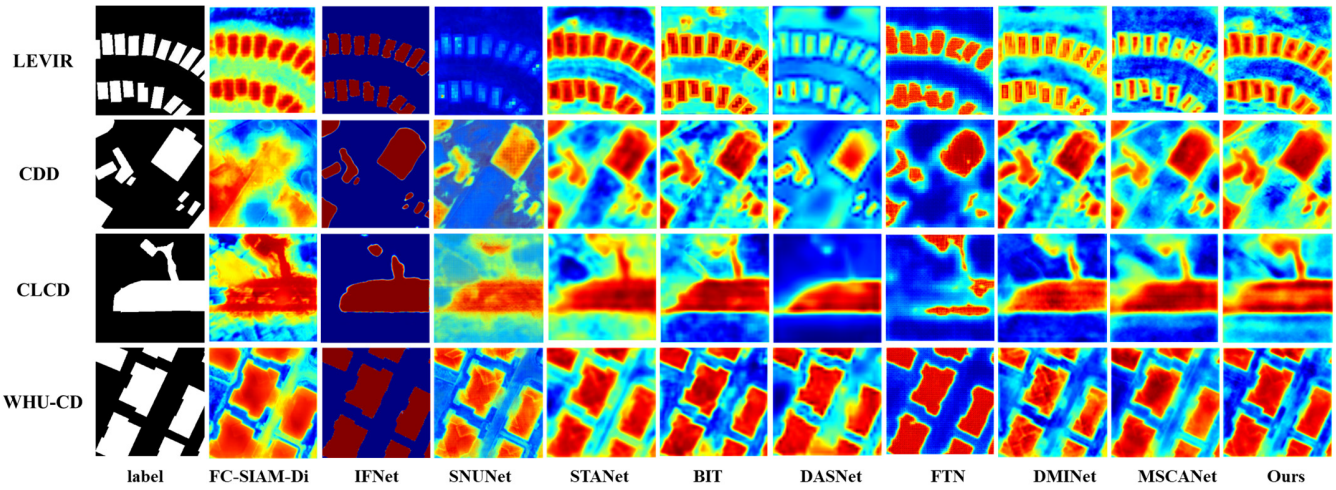

Fig. 9. Network visualization results of different models taking images from LEVIR-CD, CDD, CLCD, and WHU-CD as examples.

having relatively lower model computational complexity and space complexity compared to other models.

F. Neural Network Visualization

To explore the impact of each module in DA-IDANet during the processing, we specifically chose a pair of image instances from the LEVIR-CD. The visualization of each stage of the neural network is presented in Fig. 8. Initially, a pair of dual-temporal images is selected as input, and three different scale features are extracted by ResNet-18 in a Siamese structure. Subsequently, the extracted features undergo aggregation with contextual information through the encoding and decoding stages of the transformer. We introduce the IDAM to guide the model to focus more on relevant changing areas while suppressing irrelevant interference. The visual results in the figure clearly demonstrate that the IDAM enhances the model's attention toward changed areas, assigning lower attention weights to unchanged areas. Ultimately, the convolutional operation on the obtained change probability map reveals more accurate change localization and fine boundaries. Additionally, four representative image instances are selected from the WHU-CD, CDD, and CLCD datasets to visualize the final stage's predictions in Fig. 9. Through comparisons with other SOTA approaches, the proposed model excels in achieving more complete boundary detection while minimizing attention to unchanged areas. For large agricultural lands or dense buildings, there may be external factors that interfere with the level of concern for the model to judge the changed areas. However, the changed areas that DA-IDANet focuses on are the closest to the actual labeled region while other approaches tend to overlook some significant information. It demonstrates superior accuracy and effectiveness of DA-IDANet in CD tasks.

V. Conclusion

For the CD tasks using RS images, this article proposes a CNN-transformer model incorporating DACs and an IDAM. Specifically, we first extract three different scale features through CNN and utilize DACs for feature alignment to reduce the distribution difference between input images. Then, the transformer performs the encoding and decoding stage to aggregate context information. The proposed IDAM efficiently utilizes the interactive differential discriminant information of input images, dynamically minimizing interference from irrelevant information. We performed extensive experiments on four datasets: LEVIR-CD, CDD, CLCD, as well as WHU-CD. The results indicate that our approach achieves SOTA performance concerning accuracy and effectiveness compared to other approaches. While our proposed model is specifically designed for CD tasks, some modules within our approach may have potential applications in other domains. Future research endeavors may explore the adaptation of the proposed model to heterogeneous images.

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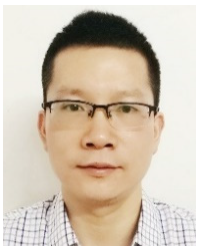

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